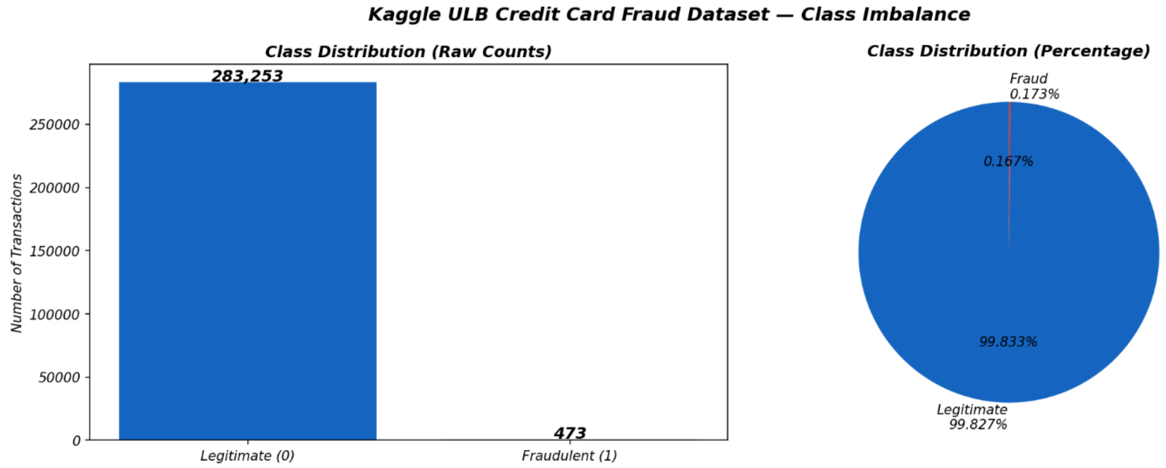
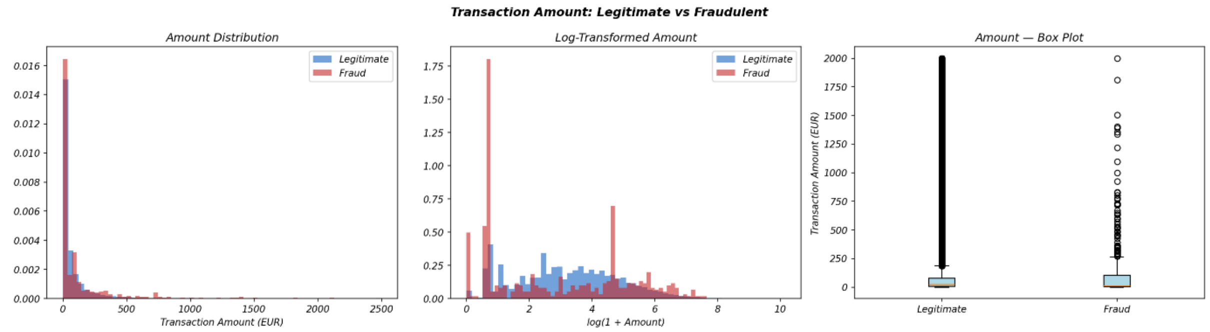
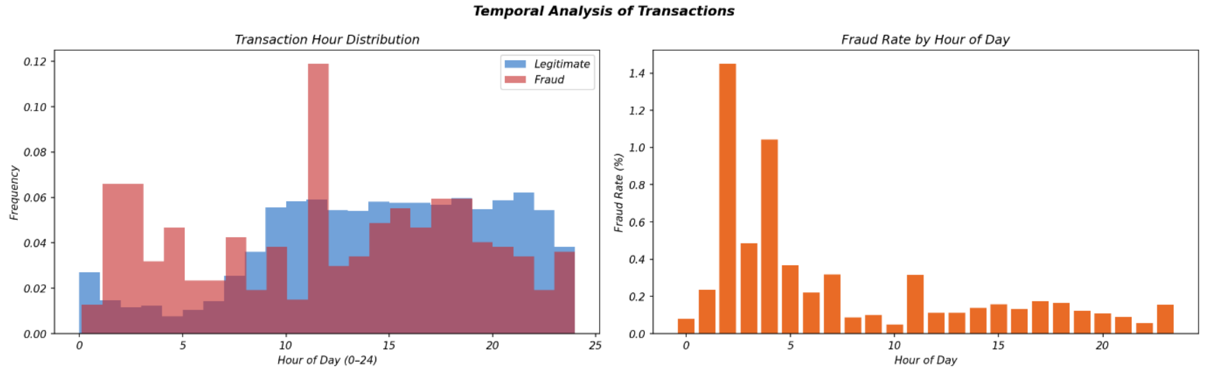
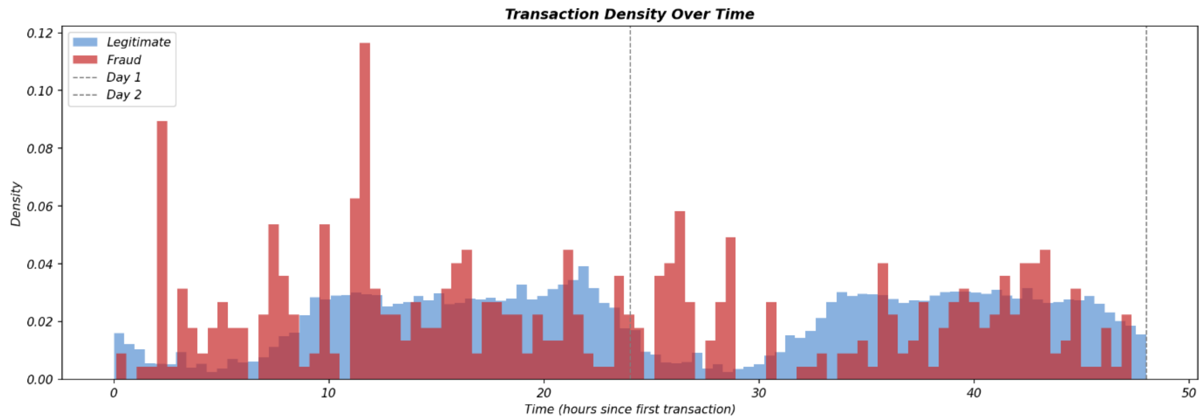
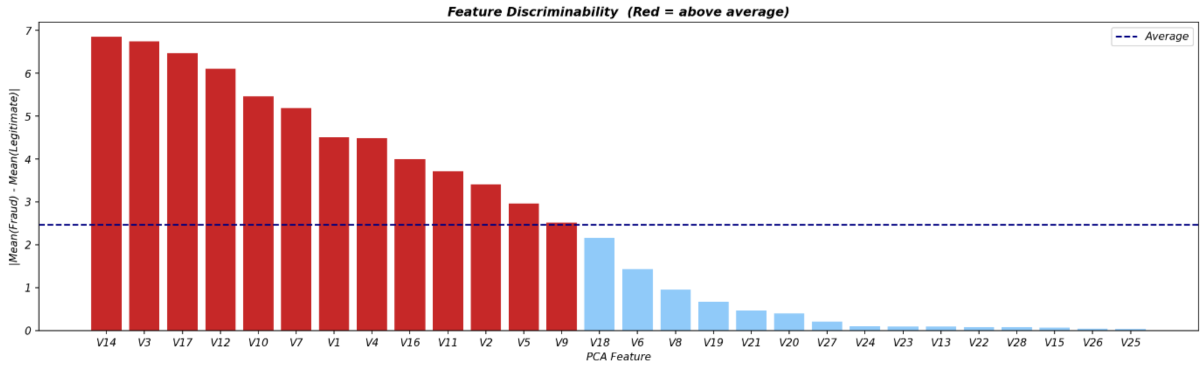
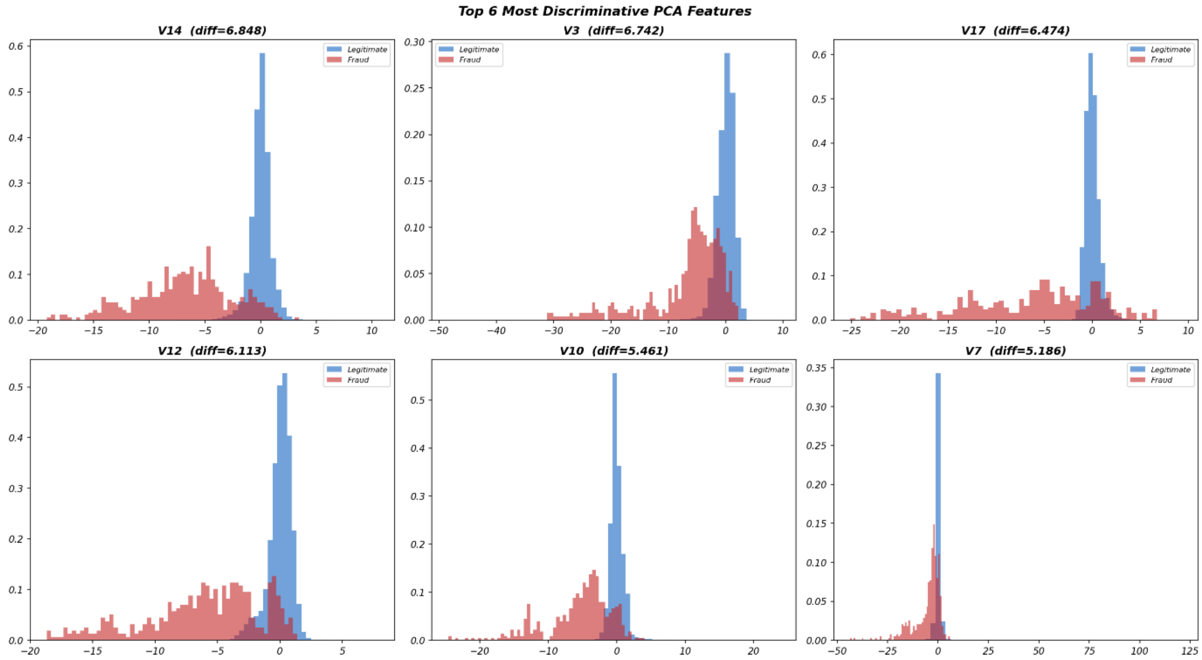

Kaggle ULB Credit Card Fraud Dataset — Class Imbalance
Class Distribution (Raw Counts)
Number of Transactions
283,253
473
Legitimate (0)
Fraudulent (1)
Class Distribution (Percentage)
Fraud
0.173%
0.167%
99.833%
Legitimate
99.827%

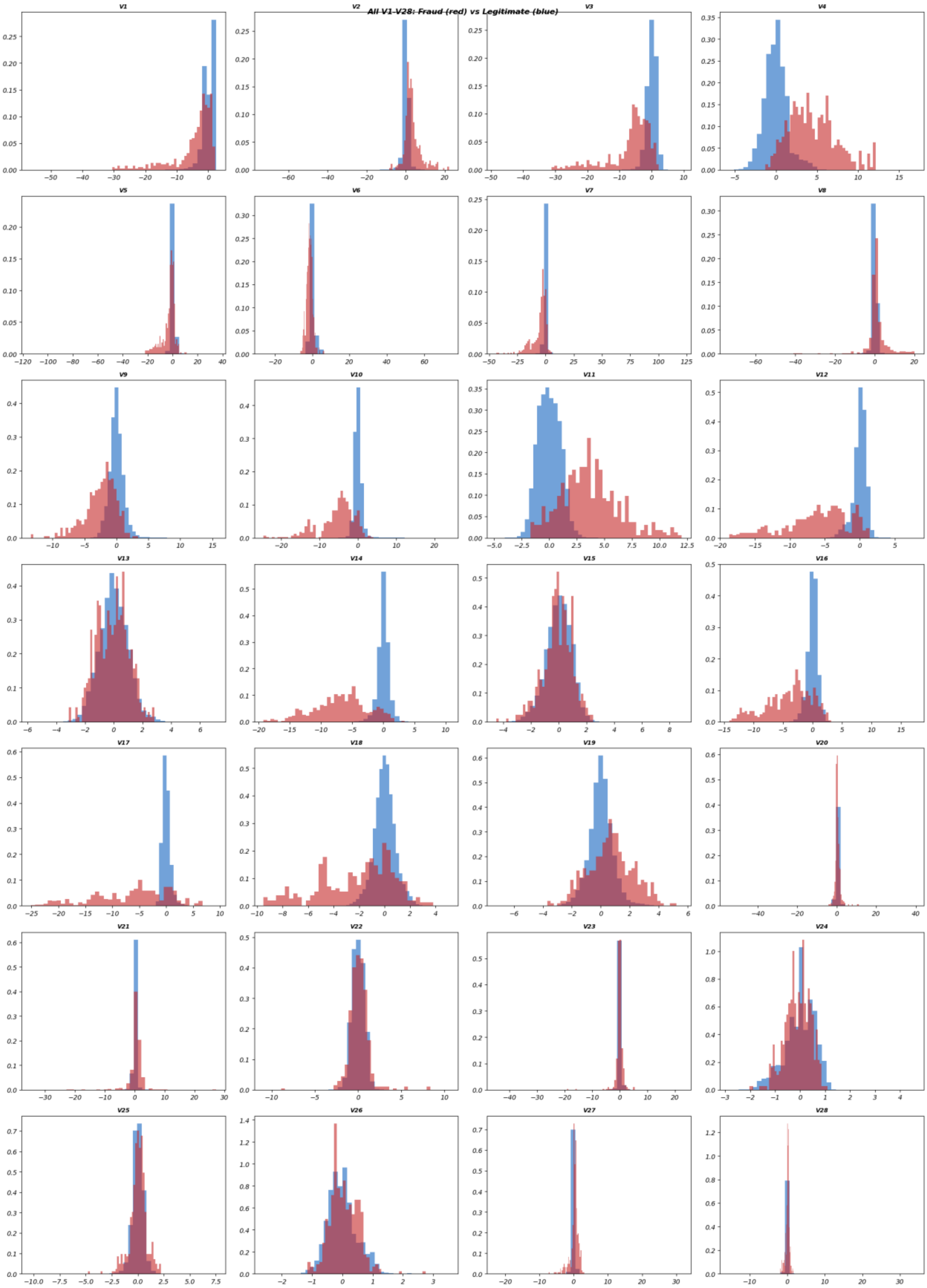

Transaction Amount: Legitimate vs Fraudulent
Amount Distribution
Legitimate
Fraud
Transaction Amount (EUR)
Log-Transformed Amount
Legitimate
Fraud
log(1 + Amount)
Amount — Box Plot
Transaction Amount (EUR)
Legitimate
Fraud

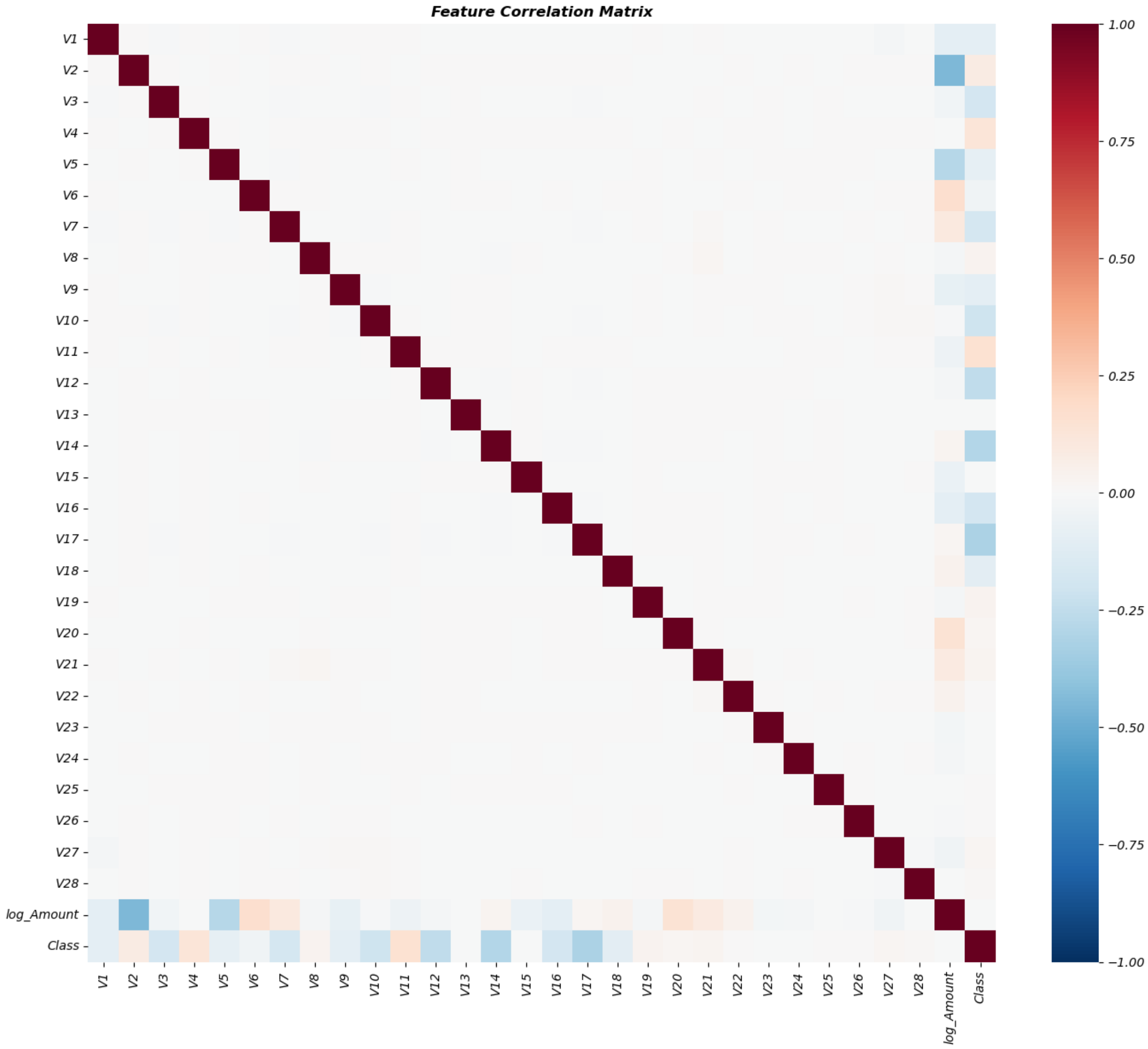

Temporal Analysis of Transactions
Transaction Hour Distribution
Legitimate
Fraud
Frequency
Hour of Day (0–24)
Fraud Rate by Hour of Day
Fraud Rate (%)
Hour of Day


Transaction Density Over Time
Legitimate
Fraud
Day 1
Day 2
Density
Time (hours since first transaction)

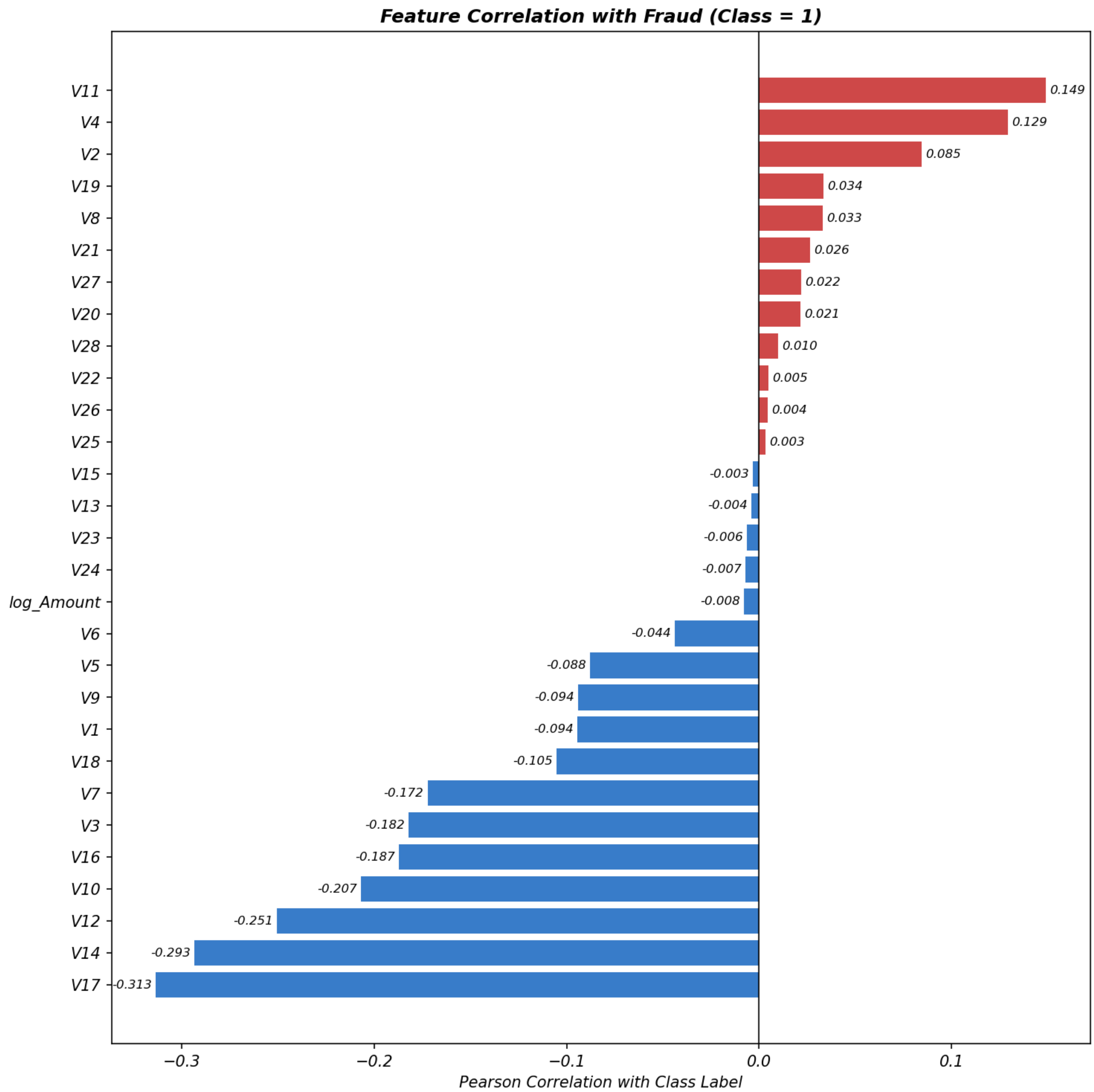

Feature Discriminability (Red = above average)
Average
|Mean(Fraud) - Mean(Legitimate)|
PCA Feature
V14
V3
V17
V12
V10
V7
V1
V4
V16
V11
V2
V5
V9
V18
V6
V8
V19
V21
V20
V27
V24
V23
V13
V22
V28
V15
V26
V25

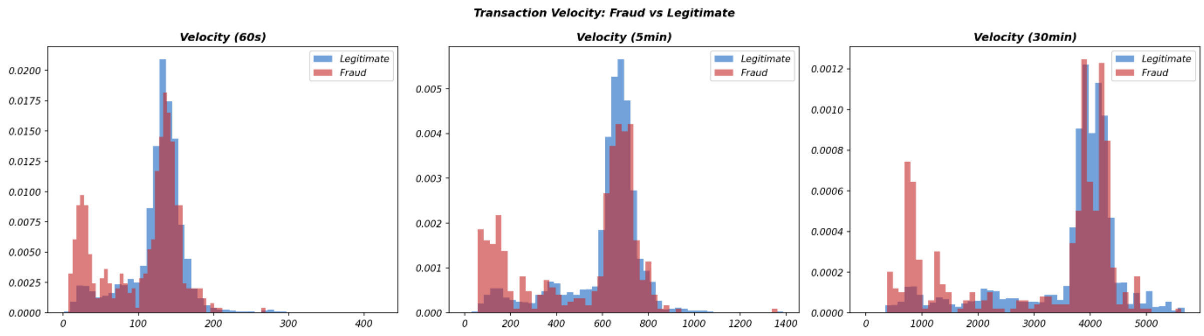

Top 6 Most Discriminative PCA Features
V14 (diff=6.848)
V3 (diff=6.742)
V17 (diff=6.474)
V12 (diff=6.113)
V10 (diff=5.461)
V7 (diff=5.186)
Legitimate
Fraud

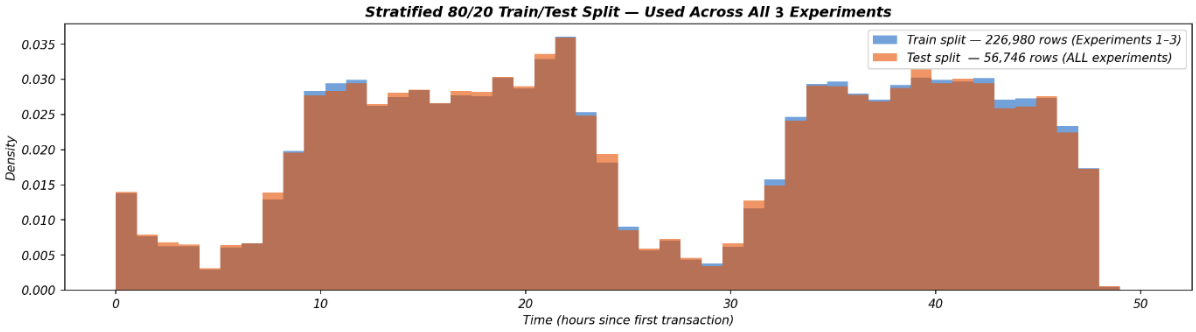

All V1-V28: Fraud (red) vs Legitimate (blue)
V1
V2
V3
V4
V5
V6
V7
V8
V9
V10
V11
V12
V13
V14
V15
V16
V17
V18
V19
V20
V21
V22
V23
V24
V25
V26
V27
V28

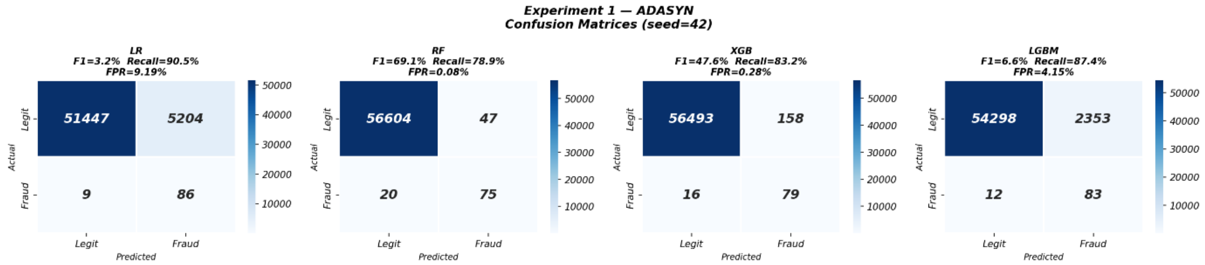


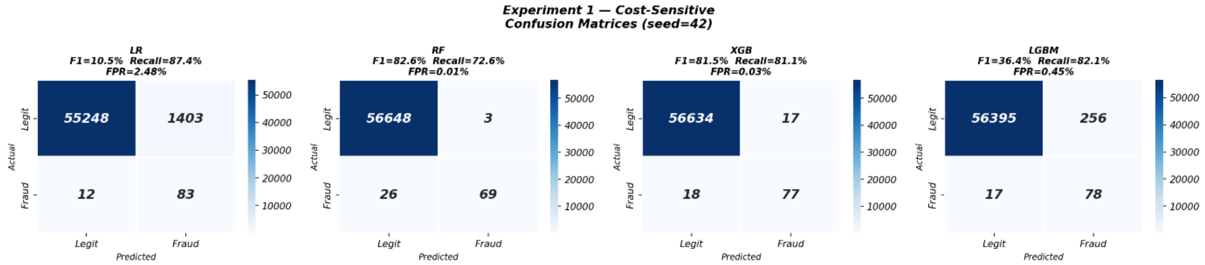
Feature Correlation Matrix
V1
V2
V3
V4
V5
V6
V7
V8
V9
V10
V11
V12
V13
V14
V15
V16
V17
V18
V19
V20
V21
V22
V23
V24
V25
V26
V27
V28
log_Amount
Class
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

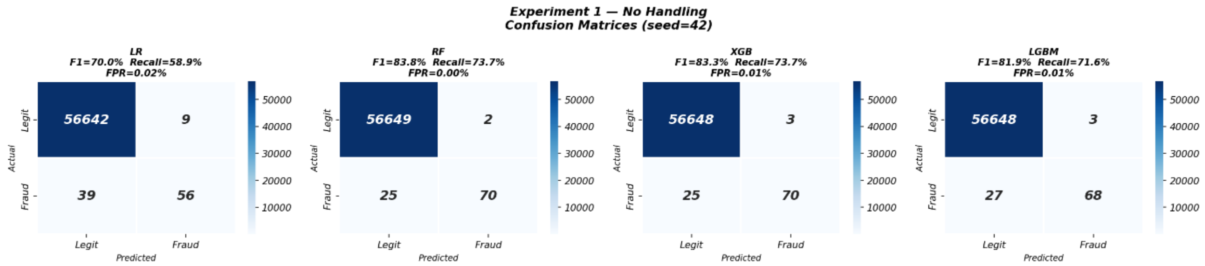

Feature Correlation with Fraud (Class = 1)
V11
V4
V2
V19
V8
V21
V27
V20
V28
V22
V26
V25
V15
V13
V23
V24
log_Amount
V6
V5
V9
V1
V18
V7
V3
V16
V10
V12
V14
V17
0.149
0.129
0.085
0.034
0.033
0.026
0.022
0.021
0.010
0.005
0.004
0.003
-0.003
-0.004
-0.006
-0.007
-0.008
-0.044
-0.088
-0.094
-0.094
-0.105
-0.172
-0.182
-0.187
-0.207
-0.251
-0.293
-0.313
−0.3
−0.2
−0.1
0.0
0.1
Pearson Correlation with Class Label

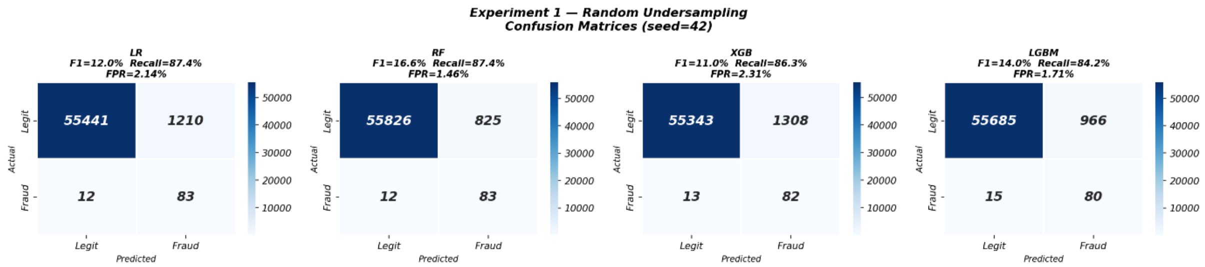

Transaction Velocity: Fraud vs Legitimate
Velocity (60s)
Velocity (5min)
Velocity (30min)
Legitimate
Fraud

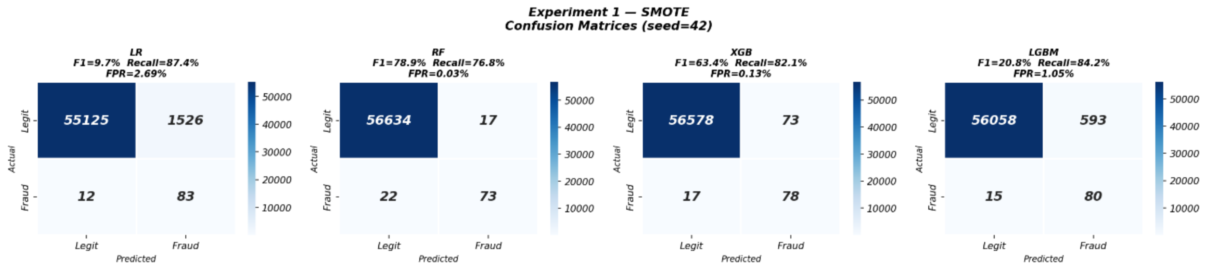

Stratified 80/20 Train/Test Split — Used Across All 3 Experiments
Train split — 226,980 rows (Experiments 1–3)
Test split — 56,746 rows (ALL experiments)
Density
Time (hours since first transaction)

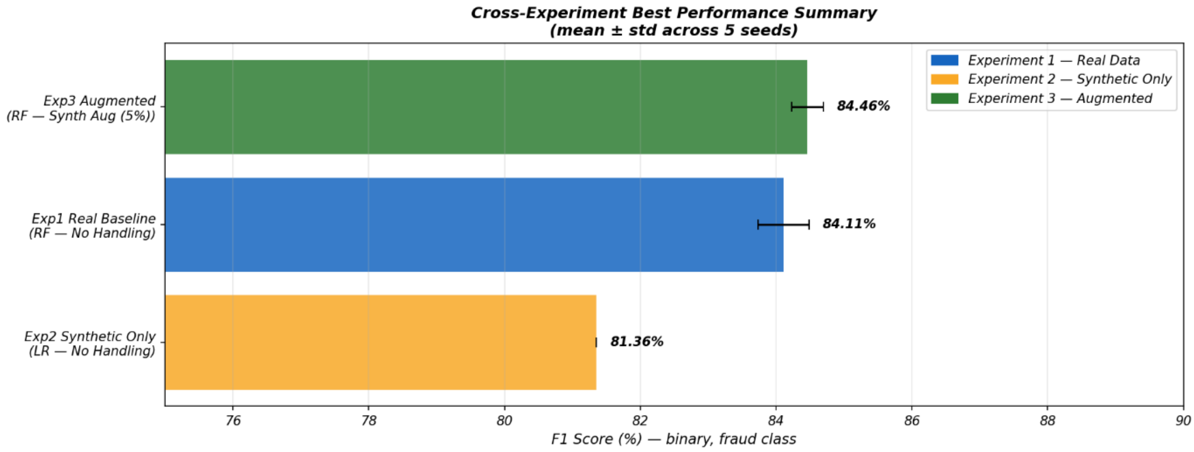

Experiment 1 — ADASYN
Confusion Matrices (seed=42)
LR
F1=3.2% Recall=90.5%
FPR=9.19%
51447
5204
9
86
RF
F1=69.1% Recall=78.9%
FPR=0.08%
56604
47
20
75
XGB
F1=47.6% Recall=83.2%
FPR=0.28%
56493
158
16
79
LGBM
F1=6.6% Recall=87.4%
FPR=4.15%
54298
2353
12
83
Actual
Predicted
Legit
Fraud

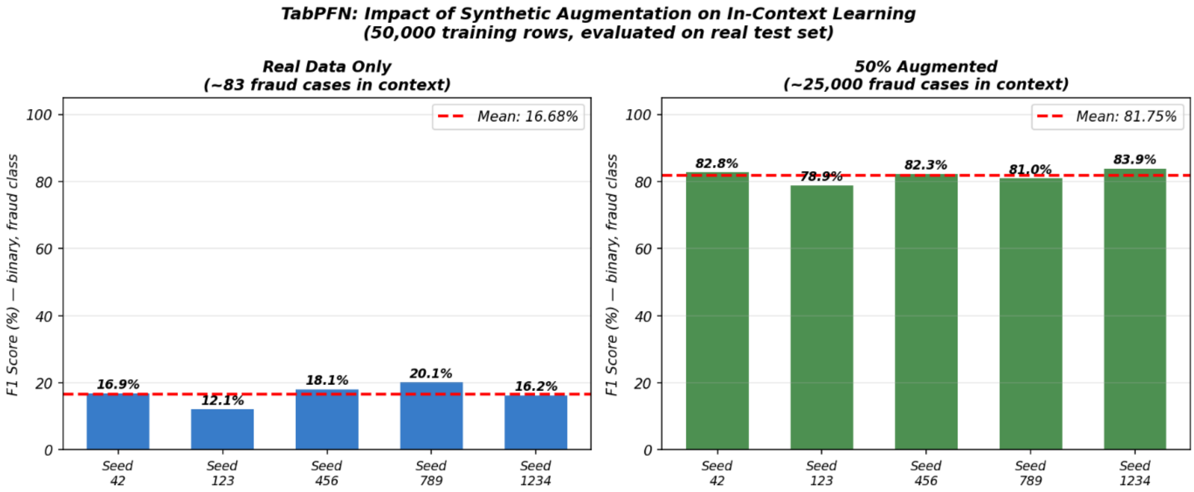

Experiment 1 — Cost-Sensitive
Confusion Matrices (seed=42)
LR
F1=10.5% Recall=87.4%
FPR=2.48%
55248
1403
12
83
RF
F1=82.6% Recall=72.6%
FPR=0.01%
56648
3
26
69
XGB
F1=81.5% Recall=81.1%
FPR=0.03%
56634
17
18
77
LGBM
F1=36.4% Recall=82.1%
FPR=0.45%
56395
256
17
78

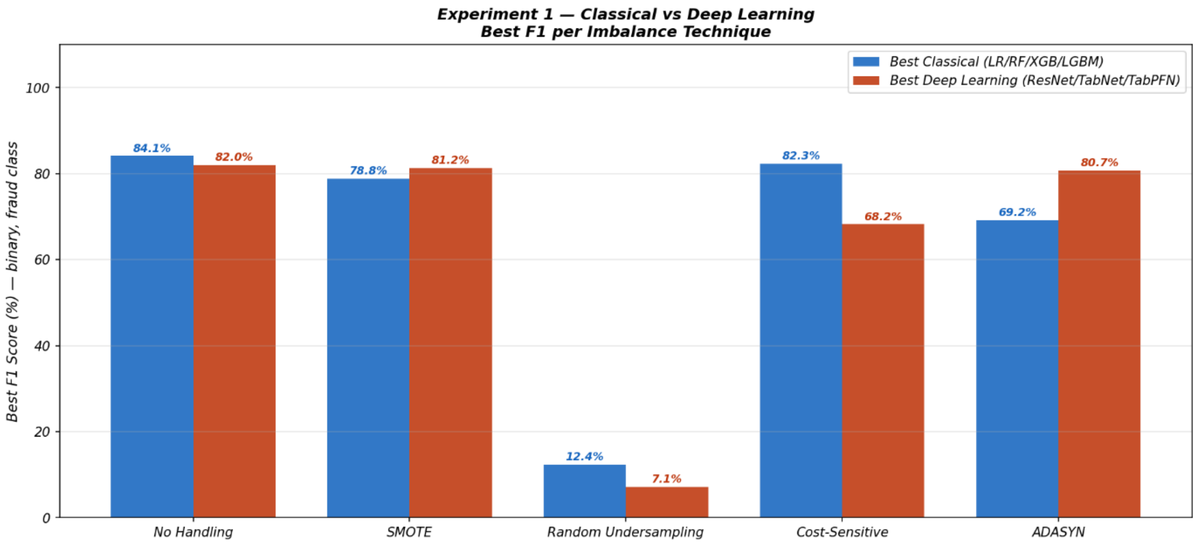

Experiment 1 — No Handling
Confusion Matrices (seed=42)
LR
F1=70.0% Recall=58.9%
FPR=0.02%
56642
9
39
56
RF
F1=83.8% Recall=73.7%
FPR=0.00%
56649
2
25
70
XGB
F1=83.3% Recall=73.7%
FPR=0.01%
56648
3
25
70
LGBM
F1=81.9% Recall=71.6%
FPR=0.01%
56648
3
27
68

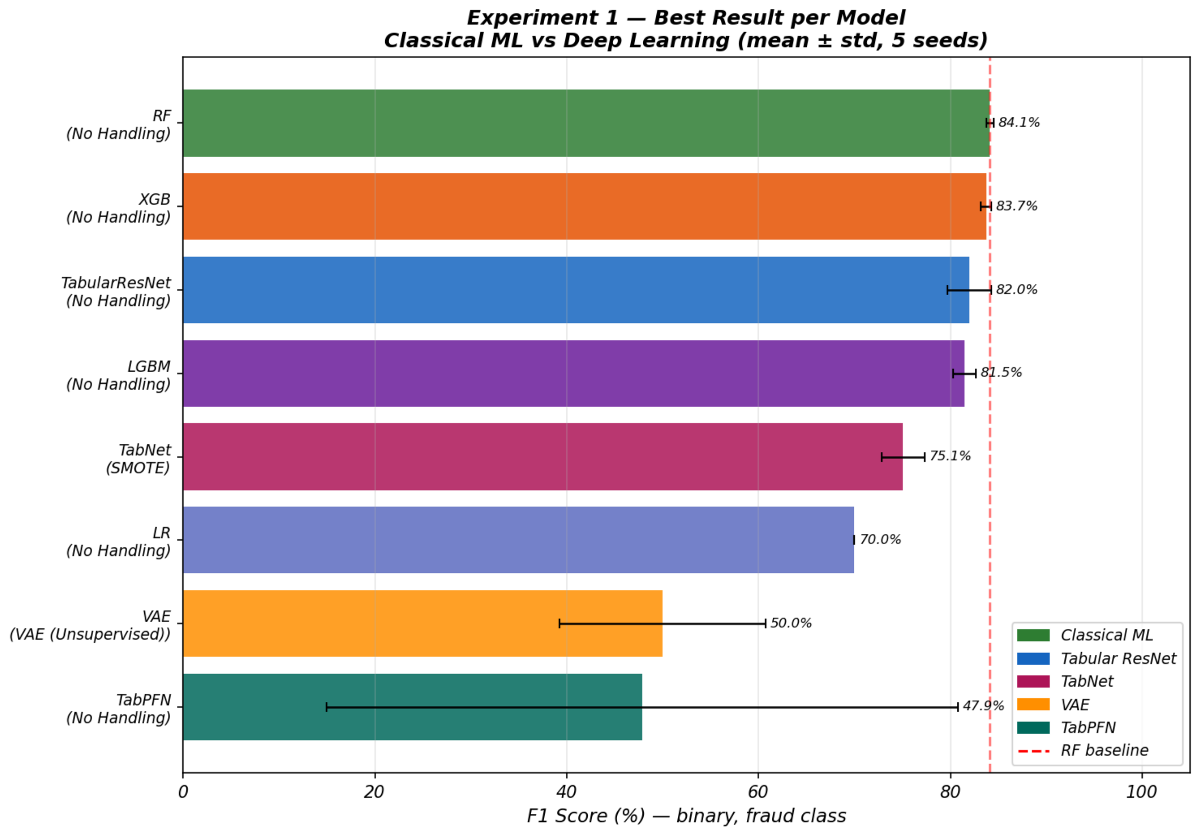

Experiment 1 — Random Undersampling
Confusion Matrices (seed=42)
LR
F1=12.0% Recall=87.4%
FPR=2.14%
55441
1210
12
83
RF
F1=16.6% Recall=87.4%
FPR=1.46%
55826
825
12
83
XGB
F1=11.0% Recall=86.3%
FPR=2.31%
55343
1308
13
82
LGBM
F1=14.0% Recall=84.2%
FPR=1.71%
55685
966
15
80

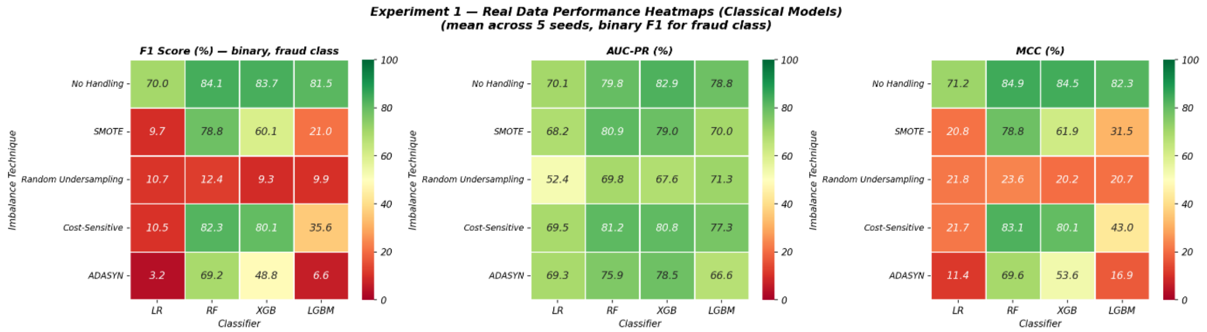

Experiment 1 — SMOTE
Confusion Matrices (seed=42)
LR
F1=9.7% Recall=87.4%
FPR=2.69%
55125
1526
12
83
RF
F1=78.9% Recall=76.8%
FPR=0.03%
56634
17
22
73
XGB
F1=63.4% Recall=82.1%
FPR=0.13%
56578
73
17
78
LGBM
F1=20.8% Recall=84.2%
FPR=1.05%
56058
593
15
80

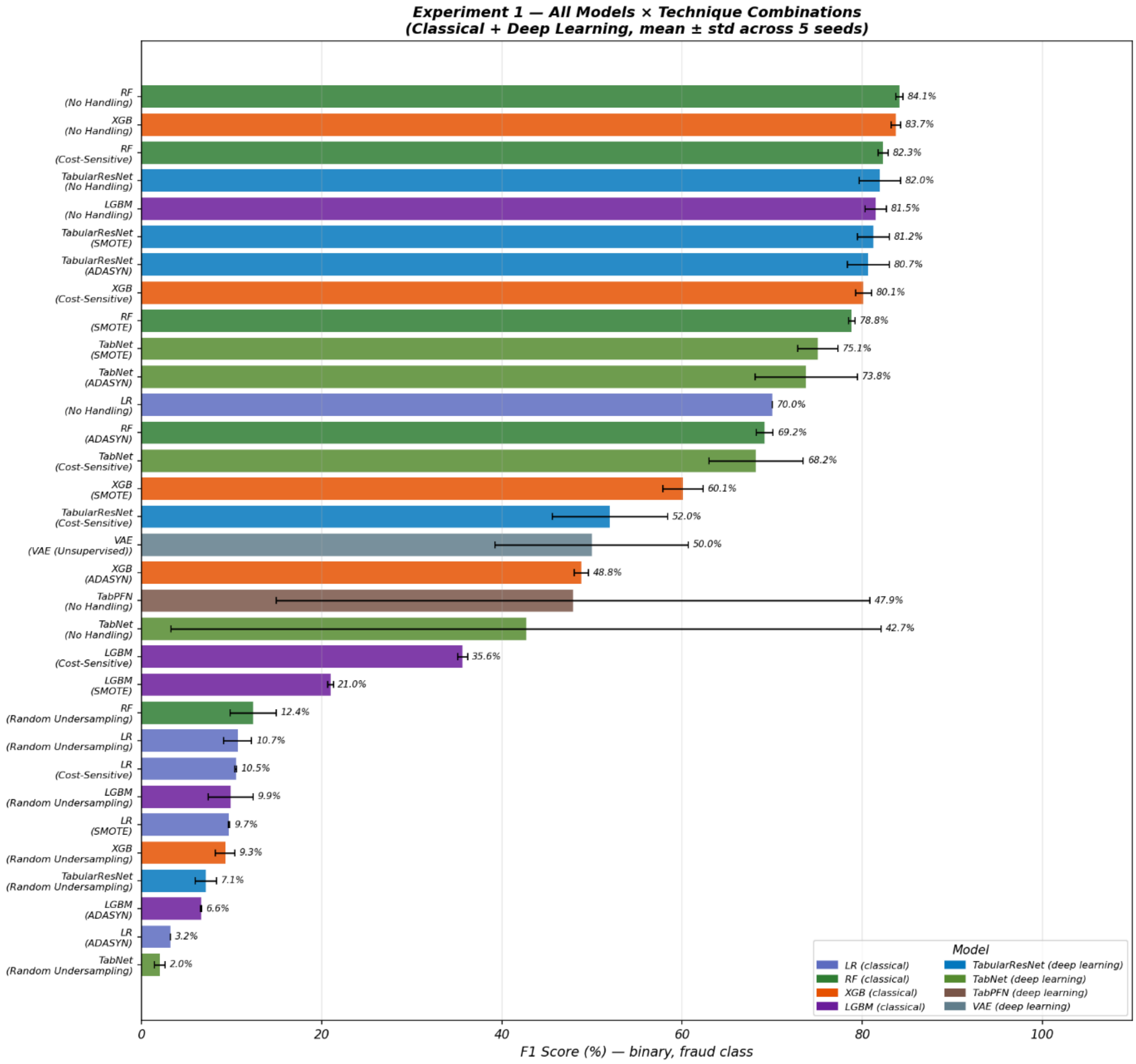

Cross-Experiment Best Performance Summary
(mean ± std across 5 seeds)
Experiment 1 — Real Data
Experiment 2 — Synthetic Only
Experiment 3 — Augmented
Exp3 Augmented
(RF — Synth Aug (5%))
Exp1 Real Baseline
(RF — No Handling)
Exp2 Synthetic Only
(LR — No Handling)
84.46%
84.11%
81.36%
76
78
80
82
84
86
88
90
F1 Score (%) — binary, fraud class

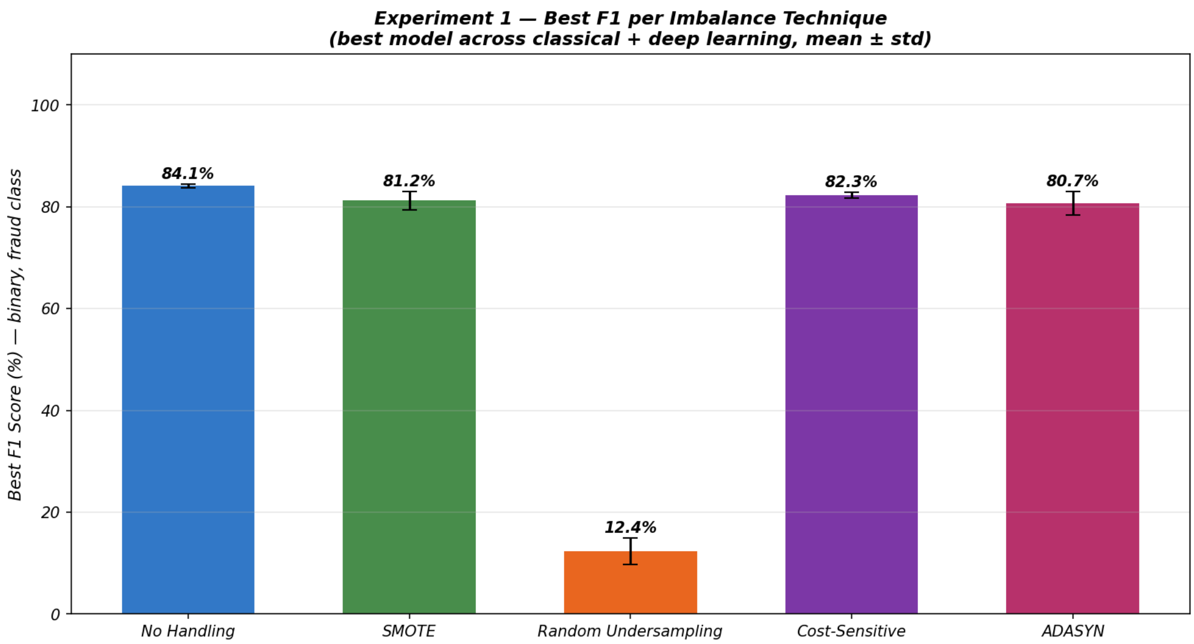

TabPFN: Impact of Synthetic Augmentation on In-Context Learning
(50,000 training rows, evaluated on real test set)
Real Data Only
(~83 fraud cases in context)
Mean: 16.68%
16.9%
12.1%
18.1%
20.1%
16.2%
50% Augmented
(~25,000 fraud cases in context)
Mean: 81.75%
82.8%
78.9%
82.3%
81.0%
83.9%
F1 Score (%) — binary, fraud class
0
20
40
60
80
100
Seed 42
Seed 123
Seed 456
Seed 789
Seed 1234

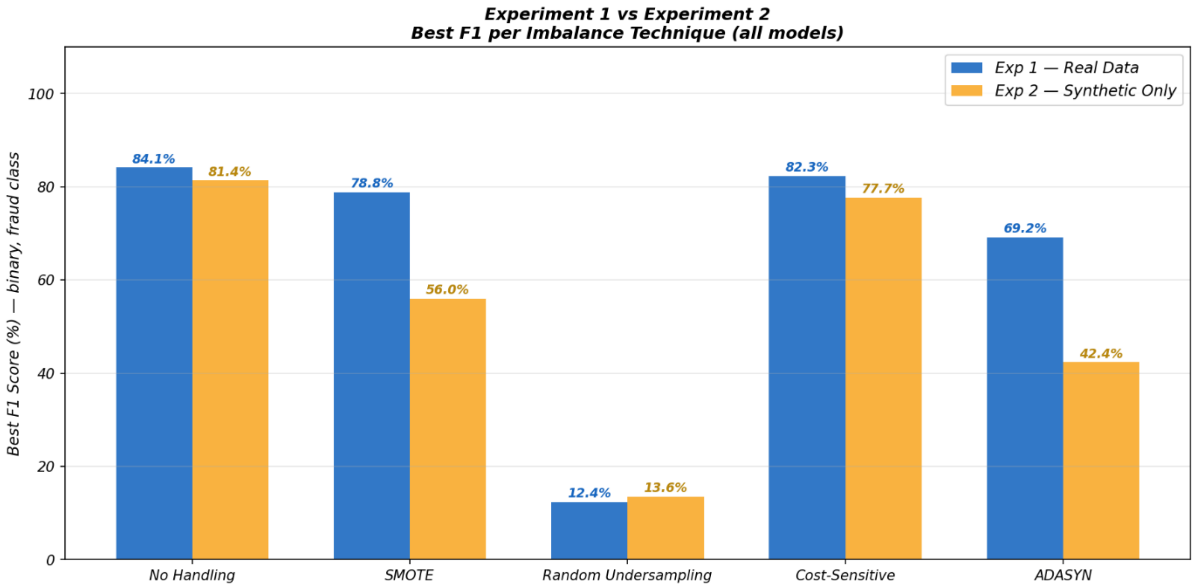

Experiment 1 — Classical vs Deep Learning
Best F1 per Imbalance Technique
Best Classical (LR/RF/XGB/LGBM)
Best Deep Learning (ResNet/TabNet/TabPFN)
84.1%
82.0%
78.8%
81.2%
12.4%
7.1%
82.3%
68.2%
69.2%
80.7%
Best F1 Score (%) — binary, fraud class
0
20
40
60
80
100
No Handling
SMOTE
Random Undersampling
Cost-Sensitive
ADASYN

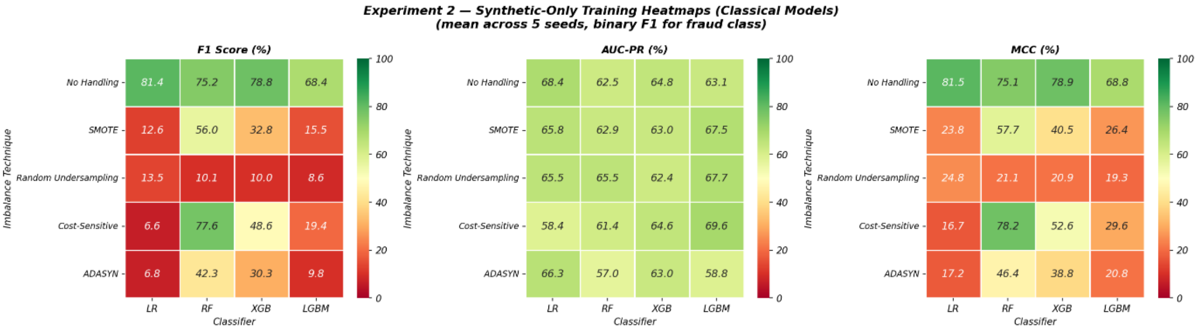

Experiment 1 — Best Result per Model
Classical ML vs Deep Learning (mean ± std, 5 seeds)
RF (No Handling)
XGB (No Handling)
TabularResNet (No Handling)
LGBM (No Handling)
TabNet (SMOTE)
LR (No Handling)
VAE (VAE (Unsupervised))
TabPFN (No Handling)
84.1%
83.7%
82.0%
81.5%
75.1%
70.0%
50.0%
47.9%
Classical ML
Tabular ResNet
TabNet
VAE
TabPFN
RF baseline
0
20
40
60
80
100
F1 Score (%) — binary, fraud class

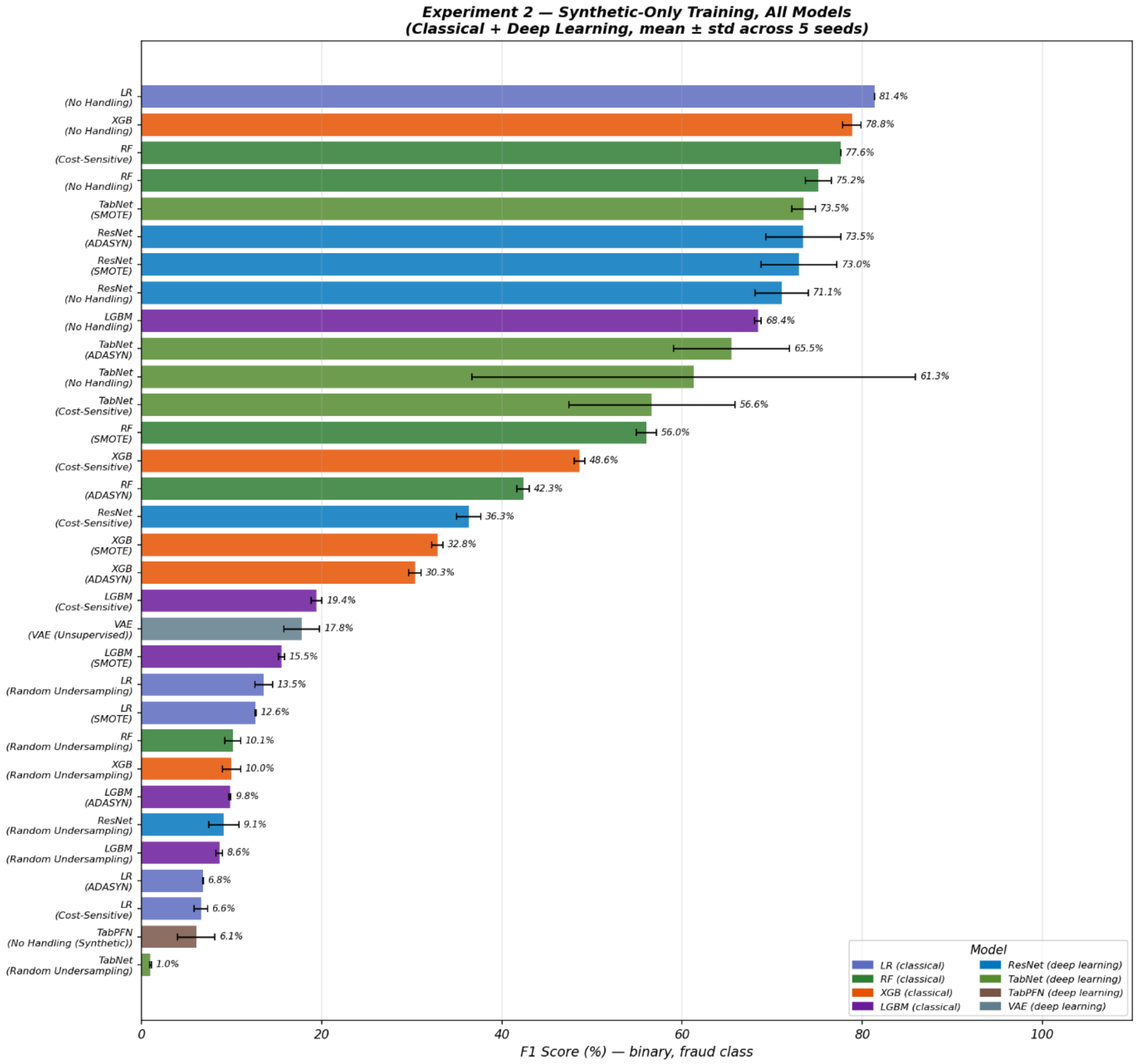

Experiment 1 — Real Data Performance Heatmaps (Classical Models)
(mean across 5 seeds, binary F1 for fraud class)
F1 Score (%) — binary, fraud class
AUC-PR (%)
MCC (%)
Imbalance Technique
No Handling
SMOTE
Random Undersampling
Cost-Sensitive
ADASYN
LR
RF
XGB
LGBM
Classifier
70.0 84.1 83.7 81.5
9.7 78.8 60.1 21.0
10.7 12.4 9.3 9.9
10.5 82.3 80.1 35.6
3.2 69.2 48.8 6.6
70.1 79.8 82.9 78.8
68.2 80.9 79.0 70.0
52.4 69.8 67.6 71.3
69.5 81.2 80.8 77.3
69.3 75.9 78.5 66.6
71.2 84.9 84.5 82.3
20.8 78.8 61.9 31.5
21.8 23.6 20.2 20.7
21.7 83.1 80.1 43.0
11.4 69.6 53.6 16.9
0
20
40
60
80
100

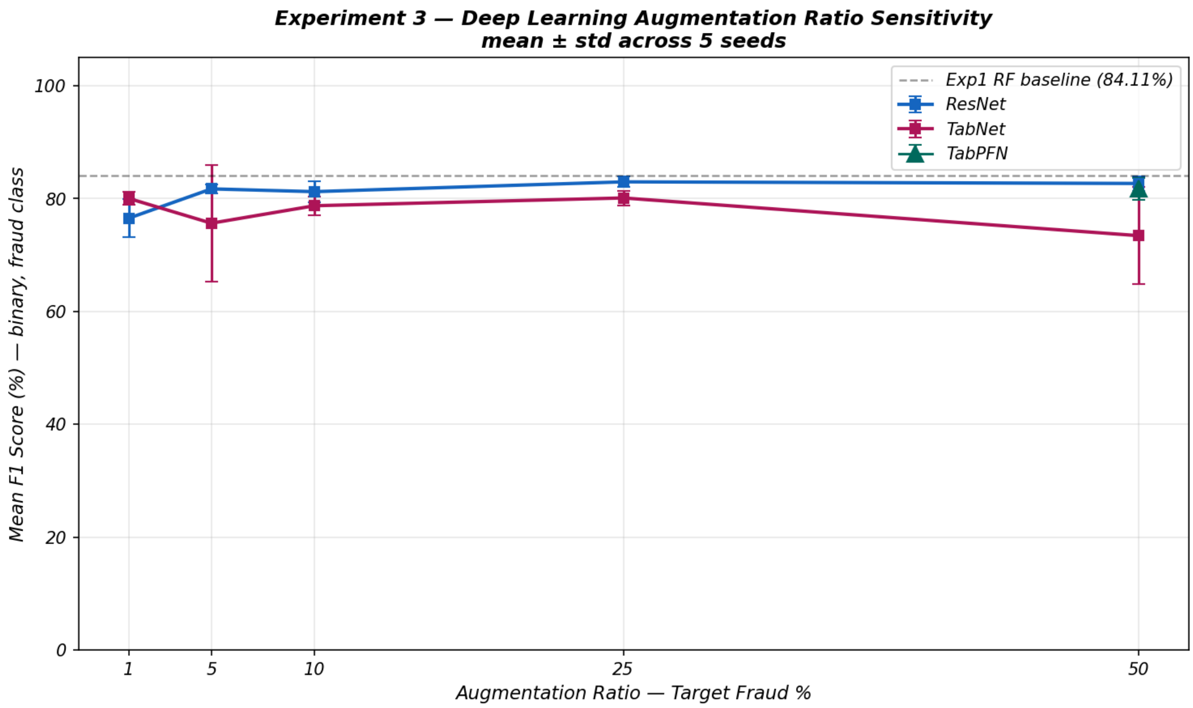

Experiment 1 — All Models × Technique Combinations
(Classical + Deep Learning, mean ± std across 5 seeds)
RF (No Handling) 84.1%
XGB (No Handling) 83.7%
RF (Cost-Sensitive) 82.3%
TabularResNet (No Handling) 82.0%
LGBM (No Handling) 81.5%
TabularResNet (SMOTE) 81.2%
TabularResNet (ADASYN) 80.7%
XGB (Cost-Sensitive) 80.1%
RF (SMOTE) 78.8%
TabNet (SMOTE) 75.1%
TabNet (ADASYN) 73.8%
LR (No Handling) 70.0%
RF (ADASYN) 69.2%
TabNet (Cost-Sensitive) 68.2%
XGB (SMOTE) 60.1%
TabularResNet (Cost-Sensitive) 52.0%
VAE (VAE (Unsupervised)) 50.0%
XGB (ADASYN) 48.8%
TabPFN (No Handling) 47.9%
TabNet (No Handling) 42.7%
LGBM (Cost-Sensitive) 35.6%
LGBM (SMOTE) 21.0%
RF (Random Undersampling) 12.4%
LR (Random Undersampling) 10.7%
LR (Cost-Sensitive) 10.5%
LGBM (Random Undersampling) 9.9%
LR (SMOTE) 9.7%
XGB (Random Undersampling) 9.3%
TabularResNet (Random Undersampling) 7.1%
LGBM (ADASYN) 6.6%
LR (ADASYN) 3.2%
TabNet (Random Undersampling) 2.0%
Model
LR (classical)
RF (classical)
XGB (classical)
LGBM (classical)
TabularResNet (deep learning)
TabNet (deep learning)
TabPFN (deep learning)
VAE (deep learning)
0
20
40
60
80
100
F1 Score (%) — binary, fraud class

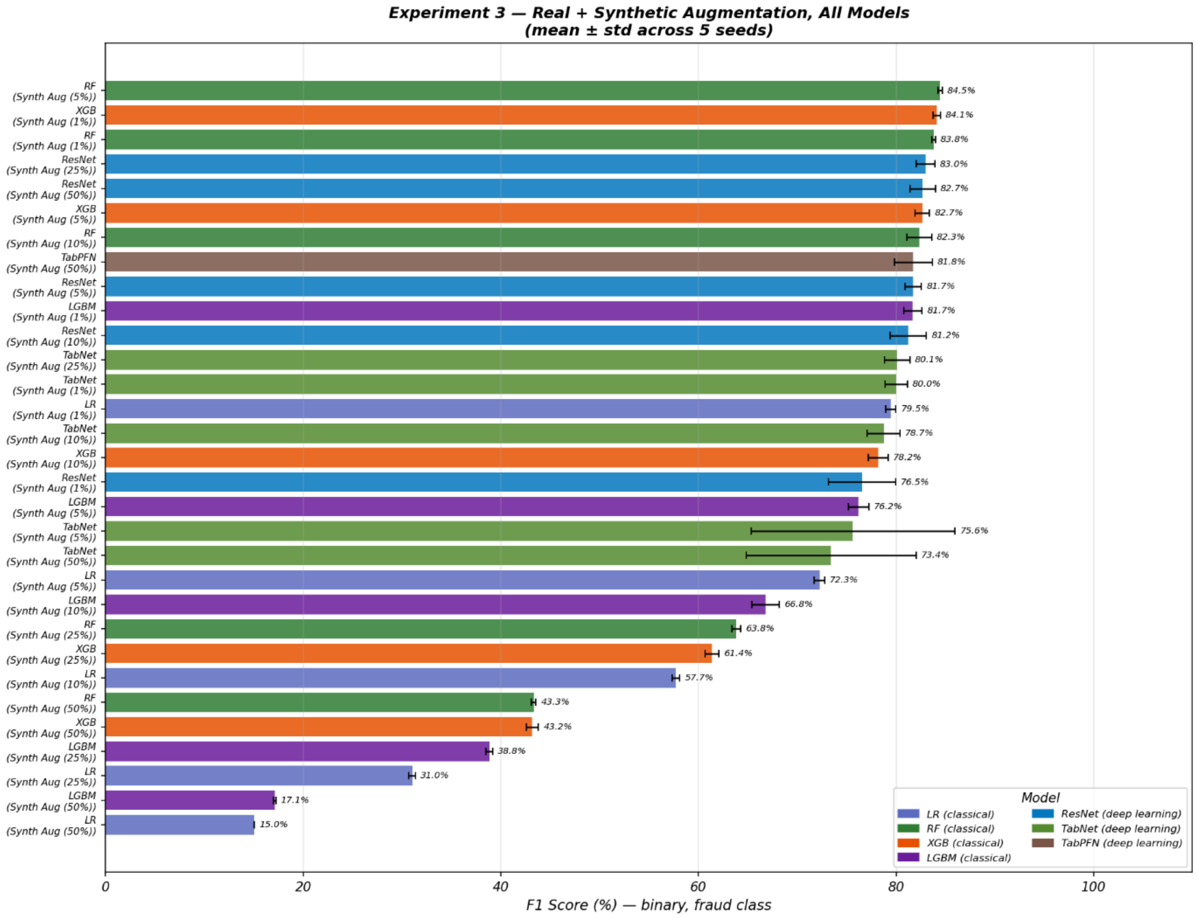

Experiment 1 — Best F1 per Imbalance Technique
(best model across classical + deep learning, mean ± std)
Best F1 Score (%) — binary, fraud class
0
20
40
60
80
100
84.1%
81.2%
12.4%
82.3%
80.7%
No Handling
SMOTE
Random Undersampling
Cost-Sensitive
ADASYN

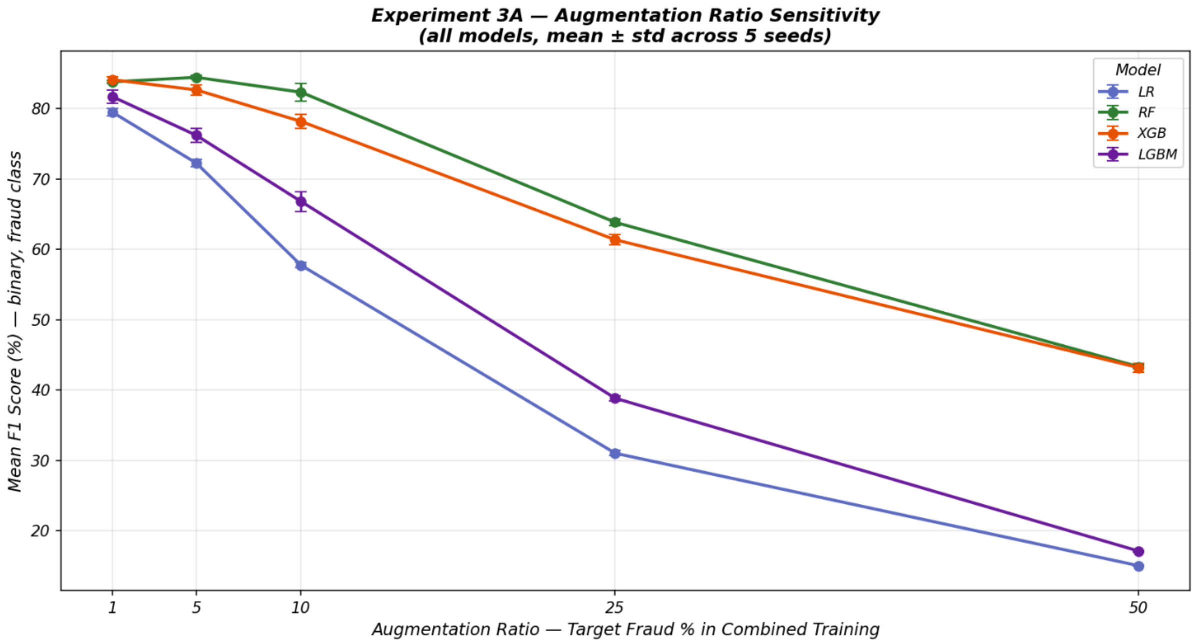

Experiment 1 vs Experiment 2
Best F1 per Imbalance Technique (all models)
Exp 1 — Real Data
Exp 2 — Synthetic Only
Best F1 Score (%) — binary, fraud class
0
20
40
60
80
100
84.1%
81.4%
78.8%
56.0%
12.4%
13.6%
82.3%
77.7%
69.2%
42.4%
No Handling
SMOTE
Random Undersampling
Cost-Sensitive
ADASYN

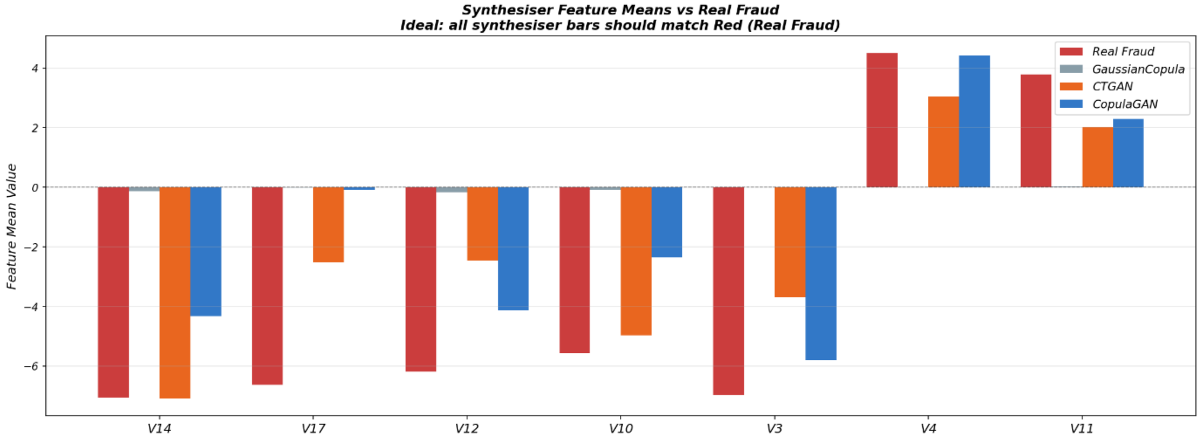

Experiment 2 — Synthetic-Only Training Heatmaps (Classical Models)
(mean across 5 seeds, binary F1 for fraud class)
F1 Score (%)
AUC-PR (%)
MCC (%)
Imbalance Technique
Classifier
LR
RF
XGB
LGBM
No Handling
SMOTE
Random Undersampling
Cost-Sensitive
ADASYN
81.4 75.2 78.8 68.4
12.6 56.0 32.8 15.5
13.5 10.1 10.0 8.6
6.6 77.6 48.6 19.4
6.8 42.3 30.3 9.8
68.4 62.5 64.8 63.1
65.8 62.9 63.0 67.5
65.5 65.5 62.4 67.7
58.4 61.4 64.6 69.6
66.3 57.0 63.0 58.8
81.5 75.1 78.9 68.8
23.8 57.7 40.5 26.4
24.8 21.1 20.9 19.3
16.7 78.2 52.6 29.6
17.2 46.4 38.8 20.8
0
20
40
60
80
100

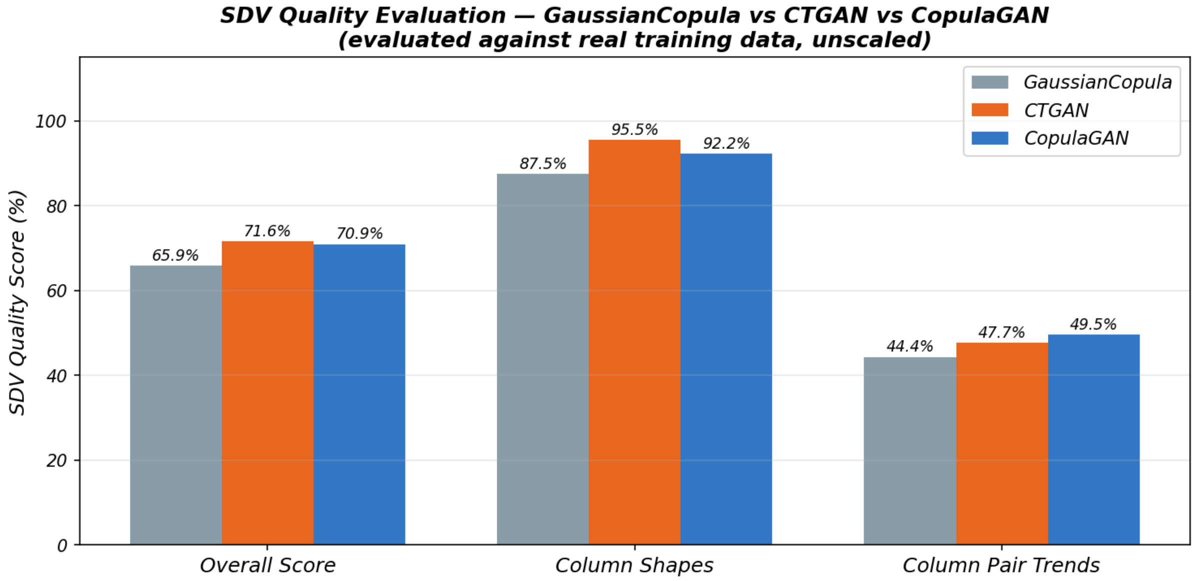

Experiment 2 — Synthetic-Only Training, All Models
(Classical + Deep Learning, mean ± std across 5 seeds)
LR (No Handling) 81.4%
XGB (No Handling) 78.8%
RF (Cost-Sensitive) 77.6%
RF (No Handling) 75.2%
TabNet (SMOTE) 73.5%
ResNet (ADASYN) 73.5%
ResNet (SMOTE) 73.0%
ResNet (No Handling) 71.1%
LGBM (No Handling) 68.4%
TabNet (ADASYN) 65.5%
TabNet (No Handling) 61.3%
TabNet (Cost-Sensitive) 56.6%
RF (SMOTE) 56.0%
XGB (Cost-Sensitive) 48.6%
RF (ADASYN) 42.3%
ResNet (Cost-Sensitive) 36.3%
XGB (SMOTE) 32.8%
XGB (ADASYN) 30.3%
LGBM (Cost-Sensitive) 19.4%
VAE (VAE (Unsupervised)) 17.8%
LGBM (SMOTE) 15.5%
LR (Random Undersampling) 13.5%
LR (SMOTE) 12.6%
RF (Random Undersampling) 10.1%
XGB (Random Undersampling) 10.0%
LGBM (ADASYN) 9.8%
ResNet (Random Undersampling) 9.1%
LGBM (Random Undersampling) 8.6%
LR (ADASYN) 6.8%
LR (Cost-Sensitive) 6.6%
TabPFN (No Handling (Synthetic)) 6.1%
TabNet (Random Undersampling) 1.0%
Model
LR (classical)
RF (classical)
XGB (classical)
LGBM (classical)
ResNet (deep learning)
TabNet (deep learning)
TabPFN (deep learning)
VAE (deep learning)
0
20
40
60
80
100
F1 Score (%) — binary, fraud class


Experiment 3 — Deep Learning Augmentation Ratio Sensitivity
mean ± std across 5 seeds
Exp1 RF baseline (84.11%)
ResNet
TabNet
TabPFN
Mean F1 Score (%) — binary, fraud class
0
20
40
60
80
100
1
5
10
25
50
Augmentation Ratio — Target Fraud %


Experiment 3 — Real + Synthetic Augmentation, All Models
(mean ± std across 5 seeds)
RF (Synth Aug (5%)) 84.5%
XGB (Synth Aug (1%)) 84.1%
RF (Synth Aug (1%)) 83.8%
ResNet (Synth Aug (25%)) 83.0%
ResNet (Synth Aug (50%)) 82.7%
XGB (Synth Aug (5%)) 82.7%
RF (Synth Aug (10%)) 82.3%
TabPFN (Synth Aug (50%)) 81.8%
ResNet (Synth Aug (5%)) 81.7%
LGBM (Synth Aug (1%)) 81.7%
ResNet (Synth Aug (10%)) 81.2%
TabNet (Synth Aug (25%)) 80.1%
TabNet (Synth Aug (1%)) 80.0%
LR (Synth Aug (1%)) 79.5%
TabNet (Synth Aug (10%)) 78.7%
XGB (Synth Aug (10%)) 78.2%
ResNet (Synth Aug (1%)) 76.5%
LGBM (Synth Aug (5%)) 76.2%
TabNet (Synth Aug (5%)) 75.6%
TabNet (Synth Aug (50%)) 73.4%
LR (Synth Aug (5%)) 72.3%
LGBM (Synth Aug (10%)) 66.8%
RF (Synth Aug (25%)) 63.8%
XGB (Synth Aug (25%)) 61.4%
LR (Synth Aug (10%)) 57.7%
RF (Synth Aug (50%)) 43.3%
XGB (Synth Aug (50%)) 43.2%
LGBM (Synth Aug (25%)) 38.8%
LR (Synth Aug (25%)) 31.0%
LGBM (Synth Aug (50%)) 17.1%
LR (Synth Aug (50%)) 15.0%
Model
LR (classical)
RF (classical)
XGB (classical)
LGBM (classical)
ResNet (deep learning)
TabNet (deep learning)
TabPFN (deep learning)
0
20
40
60
80
100
F1 Score (%) — binary, fraud class


Experiment 3A — Augmentation Ratio Sensitivity
(all models, mean ± std across 5 seeds)
Model
LR
RF
XGB
LGBM
Mean F1 Score (%) — binary, fraud class
20
30
40
50
60
70
80
1
5
10
25
50
Augmentation Ratio — Target Fraud % in Combined Training


Synthesiser Feature Means vs Real Fraud
Ideal: all synthesiser bars should match Red (Real Fraud)
Feature Mean Value
Real Fraud
GaussianCopula
CTGAN
CopulaGAN
V14
V17
V12
V10
V3
V4
V11


SDV Quality Evaluation — GaussianCopula vs CTGAN vs CopulaGAN
(evaluated against real training data, unscaled)
SDV Quality Score (%)
GaussianCopula
CTGAN
CopulaGAN
65.9%
71.6%
70.9%
87.5%
95.5%
92.2%
44.4%
47.7%
49.5%
Overall Score
Column Shapes
Column Pair Trends